

Customer Shopping Behavior Analysis

1. Project Overview

This project analyzes customer shopping behavior using transactional data collected from 3,900 customer purchases across multiple product categories. The objective of the project is to identify customer spending patterns, purchasing preferences, subscription trends, and product performance to support data-driven business decision-making.

The project combines Python for data preprocessing, PostgreSQL for SQL-based business analysis, and Power BI for interactive dashboard visualization.

2. Dataset Summary

Dataset Information

- Total Records: 3,900
- Total Columns: 18
- Dataset Type: Customer Shopping Transactions

Key Features

Customer Information

- Customer ID
- Age
- Gender
- Location
- Subscription Status

Purchase Information

- Item Purchased
- Category
- Purchase Amount
- Season
- Size
- Color

Shopping Behavior Information

- Review Rating

- Shipping Type
- Discount Applied
- Previous Purchases
- Payment Method
- Frequency of Purchases

Data Quality

- Missing values were identified in the Review Rating column.
- Duplicate records were checked and removed if necessary.
- Data consistency and formatting issues were handled during preprocessing.

Customer ID	Age	Gender	Item Purchased	Category	Purchase Amount (USD)	Location	Size	Color	Season	Review Rating	Subscription Status
1	55	Male	Blouse	Clothing	53	Kentucky	L	Gray	Winter	3.1	Yes
2	19	Male	Sweater	Clothing	64	Maine	L	Maroon	Winter	3.1	Yes
3	50	Male	Jeans	Clothing	73	Massachusetts	S	Maroon	Spring	3.1	Yes
4	21	Male	Sandals	Footwear	90	Rhode Island	M	Maroon	Spring	3.5	Yes
5	45	Male	Blouse	Clothing	49	Oregon	M	Turquoise	Spring	2.7	Yes
6	46	Male	Sneakers	Footwear	20	Wyoming	M	White	Summer	2.9	Yes
7	63	Male	Shirt	Clothing	85	Montana	M	Gray	Fall	3.2	Yes
8	27	Male	Shorts	Clothing	34	Louisiana	L	Charcoal	Winter	3.2	Yes
9	26	Male	Coat	Outerwear	97	West Virginia	L	Silver	Summer	2.6	Yes
10	57	Male	Handbag	Accessories	31	Missouri	M	Pink	Spring	4.8	Yes
11	53	Male	Shoes	Footwear	34	Arkansas	L	Purple	Fall	4.1	Yes
12	30	Male	Shorts	Clothing	68	Hawaii	S	Olive	Winter	4.9	Yes
13	61	Male	Coat	Outerwear	72	Delaware	M	Gold	Winter	4.5	Yes
14	65	Male	Dress	Clothing	51	New Hampshire	M	Violet	Spring	4.7	Yes
15	64	Male	Coat	Outerwear	53	New York	L	Teal	Winter	4.7	Yes
16	64	Male	Skirt	Clothing	81	Rhode Island	M	Teal	Winter	2.8	Yes
17	25	Male	Sunglasses	Accessories	36	Alabama	S	Gray	Spring	4.1	Yes
18	53	Male	Dress	Clothing	38	Mississippi	XL	Lavender	Winter	4.7	Yes
19	52	Male	Sweater	Clothing	48	Montana	S	Black	Summer	4.6	Yes

3. Exploratory Data Analysis and Data Preprocessing using Python

The dataset was cleaned and transformed using Python libraries such as Pandas and NumPy.

Data Loading

The dataset was imported using Pandas for analysis and manipulation.

Initial Exploration

The following functions were used:

- df.info()

- `df.describe()`
- `df.head()`
- `df.isnull().sum()`

These functions helped understand the structure, data types, and summary statistics of the dataset.

```
[6]: df.head()
```

	Customer ID	Age	Gender	Item Purchased	Category	Purchase Amount (USD)	Location	Size	Color	Season	Review Rating	Subscription Status	Shipping Type	Discount Applied	Promo Code Used	Previous Purchases
0	1	55	Male	Blouse	Clothing	53	Kentucky	L	Gray	Winter	3.1	Yes	Express	Yes	Yes	
1	2	19	Male	Sweater	Clothing	64	Maine	L	Maroon	Winter	3.1	Yes	Express	Yes	Yes	
2	3	50	Male	Jeans	Clothing	73	Massachusetts	S	Maroon	Spring	3.1	Yes	Free Shipping	Yes	Yes	
3	4	21	Male	Sandals	Footwear	90	Rhode Island	M	Maroon	Spring	3.5	Yes	Next Day Air	Yes	Yes	
4	5	45	Male	Blouse	Clothing	49	Oregon	M	Turquoise	Spring	2.7	Yes	Free Shipping	Yes	Yes	

```
[7]: df.info()
<class 'pandas.DataFrame'>
RangeIndex: 3900 entries, 0 to 3899
Data columns (total 18 columns):
#   Column                Non-Null Count  Dtype
---  -
0   Customer ID           3900 non-null   int64
1   Age                   3900 non-null   int64
...
```

```
[9]: # Checking if missing data or null values are present in the dataset
df.isnull().sum()
```

```
[9]: Customer ID      0
Age                0
Gender             0
Item Purchased     0
Category           0
Purchase Amount (USD)  0
Location           0
Size              0
Color             0
Season            0
Review Rating      37
Subscription Status 0
Shipping Type      0
Discount Applied   0
Promo Code Used    0
Previous Purchases 0
Payment Method     0
Frequency of Purchases 0
dtype: int64
```

```
[10]: # Imputing missing values in Review Rating column with the median rating of the product category
df['Review Rating'] = df.groupby('Category')['Review Rating'].transform(lambda x: x.fillna(x.median()))
```

```
[11]: df.isnull().sum()
```

```
[11]: Customer ID      0
Age                0
Gender             0
Item Purchased     0
Category           0
Purchase Amount (USD)  0
Location           0
Size              0
Color             0
Season            0
Review Rating      0
Subscription Status 0
```

Missing Value Handling

Missing values present in the Review Rating column were handled using median-based imputation according to product categories.

Column Standardization

Column names were converted into snake_case format for better readability and consistency.

Example:

Purchase Amount → purchase_amount

Review Rating → review_rating

Subscription Status → subscription_status

```
[12]: df.columns = df.columns.str.lower()
      df.columns = df.columns.str.replace(' ', '_')
      df = df.rename(columns={'purchase_amount_(usd)': 'purchase_amount'})

[13]: df.columns

[13]: Index(['customer_id', 'age', 'gender', 'item_purchased', 'category',
        'purchase_amount', 'location', 'size', 'color', 'season',
        'review_rating', 'subscription_status', 'shipping_type',
        'discount_applied', 'promo_code_used', 'previous_purchases',
        'payment_method', 'frequency_of_purchases'],
        dtype='str')

[14]: # create a new column age_group
      labels = ['Young Adult', 'Adult', 'Middle-aged', 'Senior']
      df['age_group'] = pd.qcut(df['age'], q=4, labels = labels)

[15]: df[['age', 'age_group']].head(10)

[15]:   age  age_group
0    55  Middle-aged
1    19  Young Adult
2    50  Middle-aged
3    21  Young Adult
4    45  Middle-aged
5    46  Middle-aged
6    63    Senior
7    27  Young Adult
8    26  Young Adult
9    57  Middle-aged

[16]: # create new column purchase_frequency_days
```

Feature Engineering

Additional features were created to improve analysis.

Age Group Creation

Customers were categorized into age groups:

- Young Adult
- Adult
- Middle-aged
- Senior

Purchase Frequency Days

A new feature named purchase_frequency_days was generated to represent customer purchasing frequency.

Data Consistency Validation

The relationship between discount_applied and promo_code_used was analyzed.

Since both columns contained identical information, the promo_code_used column was removed to reduce redundancy.

Database Integration

The cleaned dataset was loaded into PostgreSQL for SQL-based analysis and query execution.

After completing the SQL analysis, the processed dataset was exported as a CSV file (processed_data.csv) and used as the data source for Power BI dashboard development.

4. Data Analysis using SQL (PostgreSQL)

Several SQL queries were executed to answer important business questions.

1. Revenue by Gender

Objective:

Compare total revenue generated by male and female customers.

Insight:

Male customers generated higher overall revenue than female customers.

```
SELECT gender, SUM(purchase_amount) as Revenue
from customer_shopping
group by gender;
```

	gender text	revenue numeric
1	Female	75191
2	Male	157890

2. High-Spending Discount Users

Objective:

Identify customers who used discounts but still spent above the average purchase amount.

Insight:

Many customers continued to make high-value purchases even after applying discounts.

```
--Q2. which customer used a discount but still spend more than the average purchase amount
select customer_id, purchase_amount
from customer_shopping
where discount_applied='Yes' and purchase_amount >= (select AVG(purchase_amount) from cust
```

	customer_id bigint	purchase_amount bigint
1	2	64
2	3	73
3	4	90

3. Top 5 Products by Average Rating

Objective:

Find products with the highest customer satisfaction ratings.

Insight:

Products such as Gloves, Sandals, Boots, Hat, and Skirt received the highest average ratings.

```
select item_purchased, ROUND(AVG(review_rating::numeric),2) as "Average Product Rating"
from customer_shopping
group by item_purchased
order by avg(review_rating) desc
limit 5;
```

	item_purchased text	Average Product Rating numeric
1	Gloves	3.86
2	Sandals	3.84
3	Boots	3.82

Total rows: 5 Query complete 00:00:00.061

4. Shipping Type Comparison

Objective:

Compare average purchase amounts between Standard and Express shipping users.

Insight:

Customers selecting Express shipping showed slightly higher purchase amounts compared to Standard shipping users.

```
--Q4. compare the average purchase amounts between
select shipping_type,
ROUND(AVG(purchase_amount),2)
from customer_shopping
where shipping_type in ('Standard','Express')
group by shipping_type
```

	shipping_type text	round numeric
1	Standard	58.46
2	Express	60.48

5. Subscribers vs Non-Subscribers

Objective:

Compare customer spending behavior based on subscription status.

Metrics Analyzed:

- Total Customers
- Average Spending
- Total Revenue

Insight:

Non-subscribers generated greater total revenue because they represented a larger customer base.

```
select subscription_status,
COUNT(customer_id) as total_customer,
ROUND(AVG(purchase_amount),2) as avg_spend,
ROUND(SUM(purchase_amount),2) as total_revenue
from customer_shopping
group by subscription_status
order by total_revenue, avg_spend desc;
```

	subscription_status text	total_customer bigint	avg_spend numeric	total_revenue numeric
1	Yes	1053	59.49	62645.00
2	No	2847	59.87	170436.00

6. Discount-Dependent Products

Objective:

Identify products with the highest percentage of discounted purchases.

Insight:

Products such as Hats, Sneakers, Coats, Sweaters, and Pants showed high dependency on discount-based sales.

```
select item_purchased,  
ROUND(100* SUM(CASE WHEN discount_applied = 'Yes' THEN 1 ELSE 0 END)/COUNT(*),2) AS discount_rate  
from customer_shopping  
group by item_purchased  
order by discount_rate desc  
limit 5;
```

	item_purchased text	discount_rate numeric
1	Hat	50.00
2	Sneakers	49.00
3	Coat	49.00
Total rows: 5 Query complete 00:00:00		

7. Customer Segmentation

Objective:

Classify customers into behavioral groups.

Segments:

New Customers

Customers with only one previous purchase.

Returning Customers

Customers with 2–10 previous purchases.

Loyal Customers

Customers with more than 10 previous purchases.

Insight:

Most customers belonged to the Loyal Customer segment.

```
--Qr. segment customers into new, returning, and loyal based on their total number of previous purchases.
with customer_type as (
select customer_id, previous_purchases,
CASE
  WHEN previous_purchases = 1 THEN 'New'
  WHEN previous_purchases BETWEEN 2 AND 10 THEN 'Returning'
  ELSE 'Loyal'
END as customer_segment
from customer_shopping
)
select customer_segment, count(*) as "Number of Customers"
from customer_type
group by customer_segment
```

	customer_segment text	Number of Customers bigint
1	Loyal	3116
2	New	83
3	Returning	701

8. Top Products within Categories

Objective:

Determine the top-selling products in each category.

Categories Included:

- Accessories
- Clothing
- Footwear
- Outerwear

Insight:

Each category displayed a clear set of customer-favorite products.

```
--Qr. find the top 3 most purchased products within each category.
with item_counts as (
select category,
item_purchased,
COUNT(customer_id) as total_orders,
ROW_NUMBER() over(partition by category order by count(customer_id) DESC) as item_rank
from customer_shopping
group by category, item_purchased
)
select item_rank, category, item_purchased, total_orders
from item_counts
where item_rank <= 3;
```

	item_rank bigint	category text	item_purchased text	total_orders bigint
1	1	Accessori...	Jewelry	171
2	2	Accessori...	Sunglasses	161
3	3	Accessori...	Belt	161

9. Repeat Buyers and Subscription Status

Objective:

Evaluate whether repeat buyers are more likely to subscribe.

Insight:

Customers with higher purchase frequency showed stronger subscription adoption.

```
select subscription_status,
count(customer_id) as repeat_buyers
from customer_shopping
where previous_purchases > 5
group by subscription_status
```

	subscription_status text	repeat_buyers bigint
1	No	2518
2	Yes	958

10. Revenue Contribution by Age Group

Objective:

Measure revenue generated by different age groups.

Insight:

Young Adults contributed the highest overall revenue, followed by Middle-aged customers.

```
select age_group,
SUM(purchase_amount) as total_revenue
from customer_shopping
group by age_group
order by total_revenue desc;
```

	age_group text	total_revenue numeric
1	Young Adult	62143
2	Middle-aged	59197
3	Adult	55978

5. Dashboard Development using Power BI

The processed dataset generated through Python preprocessing was exported as a CSV file and imported directly into Power BI Desktop.

The dashboard was designed to provide interactive insights into customer purchasing behavior and business performance.

Dashboard KPIs

Total Customers

Displays the total number of customers included in the dataset.

Average Purchase Amount

Shows the average spending per purchase.

Average Review Rating

Displays overall customer satisfaction levels.

Dashboard Visualizations

Revenue by Category

Analyzes revenue generated across product categories.

Sales by Category

Shows purchase distribution among categories.

Revenue by Age Group

Displays revenue contribution from different age groups.

Sales by Age Group

Shows customer purchase activity by age segment.

Subscription Status Distribution

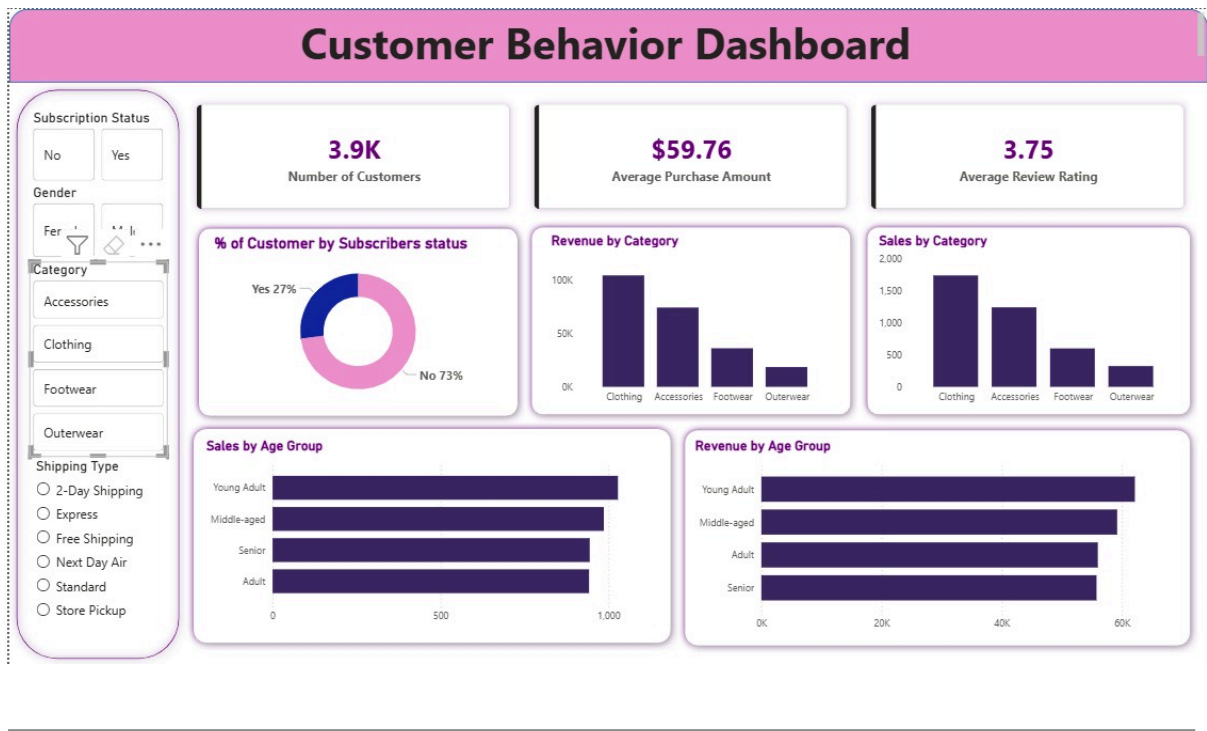
Visualizes subscriber and non-subscriber proportions.

Interactive Filters

Dashboard filters include:

- Gender
- Category
- Shipping Type
- Subscription Status

These filters allow users to dynamically explore customer behavior.



6. Business Recommendations

Increase Subscription Adoption

Provide exclusive discounts, loyalty rewards, and premium benefits to encourage more customers to subscribe.

Strengthen Customer Loyalty Programs

Reward returning customers through personalized offers and membership programs.

Optimize Discount Strategy

Focus discounts on products that demonstrate strong responsiveness to promotional campaigns.

Promote Top-Rated Products

Use highly rated products in marketing campaigns to increase customer trust and conversion rates.

Age-Based Marketing

Develop targeted campaigns for high-revenue age groups to maximize customer engagement.

Enhance Express Shipping Services

Since Express shipping customers tend to spend more, premium delivery options should be further promoted.

7. Conclusion

This project successfully analyzed customer shopping behavior using Python, PostgreSQL, and Power BI. Data preprocessing and feature engineering improved data quality and enabled deeper insights. SQL queries helped answer key business questions, while Power BI visualizations transformed the findings into an interactive dashboard.

The insights obtained from this analysis can help businesses improve customer retention, optimize marketing strategies, increase subscription adoption, and drive revenue growth through data-driven decision-making.